KEYNOTE TOOLS

User Guide & Technical Reference

Bear Engineering | AppleScript Automation Suite | v1.5

4 Programs. One Tool.

Automatically organizes photos, places arrows, and condenses slides in Keynote inspection reports.

Overview

Keynote Tools is an AppleScript program that automates the formatting of Keynote-based foundation and drainage inspection reports at Bear Engineering. It eliminates the manual, repetitive work of positioning photos, cleaning up text boxes, sizing body text, placing directional arrows, and consolidating sparse slides.

The tool contains four programs accessible from a single menu when the script is run. It is designed to be used by inspectors with no programming knowledge: launch it, pick a program, confirm the file, and let it run.

Design Intent

Inspection reports follow a consistent Keynote template: content slides each containing up to nine site photos, a body text box with observed conditions, and a slide title. Photos arrive from the field in arbitrary positions and must be repositioned into a defined grid before the report goes out.

The tool was designed around four distinct workflows:

- A full pass over a fresh report that needs everything done at once (Program 1).
- A quick re-do of a single slide after edits, with arrows (Program 2).
- A re-straightening of a single slide where the inspector has manually arranged photos and wants to preserve that order, without placing arrows (Program 3).
- A cleanup pass after content editing has left some slides partially filled (Program 4).

> *Programs 1 and 2 place directional arrows on each photo. Program 3 does not. Use Program 3 when you want to fix photo positions after already having placed arrows, so arrows are not duplicated.*

Required Files and Folder Structure

The Script

Keynote Tools.applescript is stored on the Desktop. Open it in Script Editor and click Run each time you want to use it.

The Arrows Folder

Programs 1 and 2 place PNG arrow images onto photos. These files must be stored in a subfolder called Arrows inside the same folder as the Keynote report being processed.

Required folder structure:

```
Your Job Folder/
    Foundation & Drainage Inspection Report_123 Main St.key
    Arrows/
        red_arrow_horiz.png
        red_arrow_vert.png
```

```
        blue_arrow_horiz.png
```

> *The Arrows folder must be named exactly "Arrows" (capital A) and must be a direct subfolder of the folder containing the Keynote file. If a report is moved, the Arrows folder must move with it.*

The three arrow files:

red_arrow_horiz.png	Default arrow used on most slides. Points horizontally toward the area of concern.
red_arrow_vert.png	Used on concrete cracking and rebar rusting slides. Points downward.
blue_arrow_horiz.png	Used on drainage and exterior grade/slope slides. Points horizontally in blue.

Installation for a New Inspector

1	Copy Keynote Tools.applescript to your Desktop.
2	Open it in Script Editor (search for Script Editor in Spotlight). You should see the full script text.
3	Copy the Arrows folder (containing the three PNG files) into each job folder alongside the Keynote report. Keep a master copy of the Arrows folder on your computer and duplicate it into each new job folder as needed.
4	The first time you run Program 4, macOS will ask for Accessibility permission. Go to System Settings → Privacy & Security → Accessibility and enable Script Editor. This only needs to be done once.
5	To run the tool, open Keynote Tools.applescript in Script Editor and click the Run button (▸). A menu will appear asking which program to run.

> *The script does not need to be compiled or exported as an app. Running it directly from Script Editor is the intended method.*

How to Use the Tool

Each time you run the script, a program selection menu appears. Choose the program you want and click OK.

Open File Detection

If a Keynote file is already open when you run the script, the tool will detect it and ask if you want to use that file. Click "Use This File" to proceed, or "Choose Different File" to browse for a different one.

Page Number Detection

Programs 1 and 4 auto-detect the first and last content slides by finding the slides containing the 200×267px property overview photo. Programs 2 and 3 default to whichever slide is currently visible in Keynote. Hidden slides are excluded from detection. The program runs immediately using the detected values without asking for confirmation.

The Four Programs

Program 1 — Organize All Slides (with arrows)

Run this on a fresh report before sending it out. It processes every slide in the auto-detected range and performs three passes in sequence:

Pass 1: Photo Layout

Photos are collected from each slide and placed into a standardized grid. The grid adapts to the number and type of photos on the slide:

1–3 photos	Photos placed in the top row (portrait grid, positions 1–3).
4–6 photos	Photos fill the top two rows.
7–9 photos	Photos fill all three rows.
Landscape photos (333×250px)	All photos switch to a two-column landscape grid.
Wide photos (434×267px)	Skip column 3 positions so they have room to display correctly.

Pass 2: Text Box Cleanup and Repositioning

After photos are placed, the script deletes stale Item Image label text boxes that no longer have a photo above them, deletes stray text boxes below y=880, and moves and resizes the main body text box to fill the space below the photos. On rebar rusting slides the body text box is set to 535px wide to leave room for the vertical arrow. When a wide photo is present the body text box is always placed full-width at the start of the next row down.

Pass 3: Arrow Placement

An arrow is placed centered on each photo. The arrow type is chosen based on the slide title:

Drainage / slope slides	Blue horizontal arrow.
Rebar rusting slides	Red vertical arrow (points downward from center of photo).
All other slides	Red horizontal arrow (default).

Program 2 — Fix One Slide (reset photo order, with arrows)

Performs the same photo layout, text box cleanup, and arrow placement as Program 1, but on a single slide only. Photos are reassigned sequentially to grid positions. The default slide is whichever slide is currently open in Keynote.

> *This program places new arrows on every photo. If arrows were already placed on the slide, remove them first to avoid duplicates.*

Program 3 — Fix One Slide (keep photo order, no arrows)

Performs photo layout and text box cleanup on a single slide without placing arrows. Each photo snaps to its nearest valid grid position rather than being reassigned sequentially, so any intentional manual arrangement is preserved. Use this after manually tweaking photo positions when you want the grid tidied without disturbing your ordering or adding arrows.

Program 4 — Condense Slides

Scans the auto-detected slide range top-to-bottom and merges any slide whose content fits onto the previous slide. For each slide, the script measures the total content height and the bottom edge of the previous slide. If the content fits below with a 10pt overlap gap, it shifts everything down and pastes it onto the previous slide, then deletes the now-empty slide.

Page numbers (x > 700, y > 950) are excluded from the merge and any duplicate page numbers created by the paste are deleted automatically.

> *Program 4 uses clipboard keystrokes (Cmd+A, Cmd+C, Cmd+V) to copy content between slides. Keynote must remain the frontmost application while it runs. Do not switch to another app until it finishes.*

Photo Size Reference

The script identifies photo types by their pixel dimensions in Keynote. These sizes must match exactly for the layout logic to work correctly.

Size	Type	How it's used
`200 × 267 px`	Property overview photo	Marks the first and last content slides for auto-detection of slide range.
`333 × 250 px`	Landscape site photo	Triggers the landscape grid layout for the whole slide.
`434 × 267 px`	Wide site photo	Skips column 3 in the portrait grid; forces full-width body text box placement.
`All others > 70px`	Standard site photos	Placed sequentially in the portrait grid.

Grid Position Reference

Portrait Grid (standard photos)

Row	Col 1	Col 2	Col 3
Row 1	(50, 57)	(284, 57)	(518, 57)
Row 2	(50, 357)	(284, 357)	(518, 357)
Row 3	(50, 657)	(284, 657)	(518, 657)

Landscape Grid (landscape photos)

Row	Col 1	Col 2
Row 1	(41, 73)	(394, 73)
Row 2	(41, 373)	(394, 373)
Row 3	(41, 673)	(394, 673)

Body Text Box Sizing

The body text box is identified as the text box with more than 100 characters. Its position and width are set based on photo count after layout. When a wide photo is present, the normal side-placement logic is bypassed entirely.

Photo count	Width (normal)	Width (rebar)	Position
Wide photo present (any count)	700 px	700 px	x=34, y=next row + 20pt
0, 3, or 6	700 px	535 px	x=34, y=next row start
1, 4, or 7 (short text)	465 px	465 px	x=269, y=bottom row start
1, 4, or 7 (long text >2000 chars)	700 px	535 px	x=34, y=next row start
2, 5, or 8 (short text)	232 px	232 px	x=502, y=bottom row start
2, 5, or 8 (long text >650 chars)	700 px	535 px	x=34, y=next row start

Slide Title Detection

The arrow type and body text width are determined by reading the slide title. The script identifies the title as the text box whose top edge (y position) is less than 50 points from the top of the slide canvas.

Two specific title strings trigger special behavior:

"Observed Condition - Drainage Improvements - Exterior grade/slope"	Blue horizontal arrow is used instead of red.
"Observed Condition - Concrete cracking/spalling from steel rebar rusting"	Red vertical arrow is used, and body text box width is reduced to 535px.

Title matching uses "contains" rather than an exact match, so minor variations in surrounding text will not break detection as long as the key phrase is present.

Known Limitations

- Program 4 requires Keynote to remain frontmost throughout its run, as it uses keyboard shortcuts to copy and paste content between slides. Switching to another application mid-run will cause it to fail or paste into the wrong place.
- Arrow images are placed but not scaled by the script. If the arrow PNG files are resized externally, placement positions may shift.
- The rebar rusting slide title must contain the exact phrase shown above for the narrower body text width and vertical arrow to activate.
- Hidden slides are excluded from auto-detection of start and end pages but are not excluded from Program 1 and 4 processing once the range is confirmed.

Distributing to Other Inspectors

The script is shared as a plain .applescript file and run from Script Editor. Exporting as a compiled app or Automator application introduces macOS permission issues on newer systems and is not recommended.

1	Send them Keynote Tools.applescript.
2	Send them the Arrows folder with all three PNG files.
3	Tell them to save the script to their Desktop, open it in Script Editor, and click Run each time they want to use it.
4	Have them place a copy of the Arrows folder inside each job folder alongside the Keynote report.

Version History

v1.0	Initial release with Programs 1, 2, and 3. Portrait and landscape photo grid layout, text box cleanup and repositioning based on photo count, and arrow placement with three arrow types selected by slide title.
v1.1	Added Program 4 (Condense Slides), merging sparse slide content onto the previous slide using clipboard keystrokes.

v1.2	Added open file detection across all four programs. Fixed dialog focus so page number prompts appear in front of Keynote. Keynote is brought back to front after dialogs so the run can be watched live.
v1.3	Added rebar rusting slide detection. Reworked Condense Slides to run top-to-bottom with a -10pt overlap gap. Added auto-detection of start and end pages from 200×267px property overview photos. Hidden slides excluded from detection. Current slide used as default for Programs 2 and 3.
v1.4	Initial release to BEAR team. Program 2 updated to include arrow placement. All page and slide number dialogs removed in favor of auto-detected values. Fixed off-canvas image exclusion across all three programs. Added page range auto-detection to Program 4.
v1.5	Resolved a bug related to wide photo and text box placement. When a wide photo is present, the text box is now always placed full-width (700pt) at x=34 in the next row down with 20pt additional spacing to account for the caption label.